Math 290: Gestalten

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Fall 2026

Contents

How strange to actually have to see the path of your journey in order to make it.

—Neal Shusterman, [Shu16]

Contents 2

1 Long Live the New Rings 3
- 1.1 September 3 . . . 3
  - 1.1.1 Duality . . . 3
  - 1.1.2 Sheaves as Rings . . . 4
  - 1.1.3 Going Up . . . 5
  - 1.1.4 Stefanich Rings . . . 6
- 1.2 September 10 . . . 7
  - 1.2.1 Presentable Categories . . . 7
  - 1.2.2 A Category of Categories . . . 8
  - 1.2.3 Algebra with Higher Categories . . . 9
- 1.3 September 17 . . . 10
  - 1.3.1 Tensor Products . . . 10
  - 1.3.2 Dualizable Stable Presentable Categories . . . 12
  - 1.3.3 Dualizable $\mathcal{V}$-Linear Categories . . . 15
- 1.4 October 1 . . . 16
  - 1.4.1 Stefanich Rings . . . 16
  - 1.4.2 Fun with $\mathbb{E}_\infty$-Rings . . . 17
  - 1.4.3 Descent . . . 19
- 1.5 October 8 . . . 21
  - 1.5.1 Six Functors for Stacks . . . 21
  - 1.5.2 Applications . . . 23

Bibliography 25

List of Definitions 26

Theme 1
Long Live the New Rings

1.1 September 3

After today, all classes will meet at 3:30PM.

1.1.1 Duality

We will discuss the new theory of Gestalten, due to Scholze and Stefanich. The basic motivating idea is the duality between geometry and algebra.

Example 1.1 (Gelfand)**.** Let X be a compact Hausdorff space. Then there is a commutative C^*-algebra $C(X)$ of continuous maps $X \to \mathbb{C}$. The association $X \mapsto C(X)$ induces an anti-equivalence of categories between compact Hausdorff spaces and commutative C^*-algebras.

Example 1.2 (Stone)**.** Let X be a profinite set. Then let $C_p(X)$ denote the space of continuous maps $C \to \mathbb{F}_p$. The association $X \mapsto C_p(X)$ induces an anti-equivalence of categories between profinite sets and commutative rings satisfying $x^p = x$ for all x.

Example 1.3 (Grothendieck)**.** The functor Spec from commutative rings to affine schemes is an anti-equivalence.

The last example is not exactly a theorem but instead a definition of an affine scheme. It is desirable to expand the category of geometric objects (e.g., to be able to compactify), but then we lose the equivalence. The traditional way to do this expansion is to glue affine schemes together in specified ways; in particular, the category of schemes is generated by affine schemes by taking colimits, so it embeds into $\mathrm{PSh}(\mathrm{AffSch})$. To have some geometry, we restrict from presheaves to sheaves with respect to some topology.

Remark 1.4. Similar stories of gluing is present in other areas of mathematics, such as rigid geometry, where one starts with Huber rings instead of rings.

Remark 1.5. It is desirable to work with more general notions of sheaf. In particular, to build some moduli problems, one wants to consider sheaves valued in groupoids, and for homotopy-theoretic reasons, it is convenient to work with ∞-groupoids (or spaces). Lurie has explained what categories of sheaves valued in ∞-groupoids look like, and they are called ∞-topoi. In general, one hopes that the best categories of geometric objects should be ∞-topoi.

1.1.2 Sheaves as Rings

Wherever we have geometric objects, we hope to have quasicoherent sheaves on those objects.

Example 1.6. On an affine scheme $\operatorname{Spec} R$, its category of quasicoherent sheaves is classically $\operatorname{Mod}_R^\heartsuit$; the derived category will be denoted Mod_R.

Now, we have to glue.

Example 1.7. On a scheme X, a quasicoherent sheaf M should naturally induce a module f^*M for each map $f\colon \operatorname{Spec} R \to X$. Gluing all these modules together should produce a sheaf on X, which motivates the definition

$$\operatorname{QCoh}(X) \coloneqq \lim_{R\to X} \operatorname{QCoh}(\operatorname{Spec} R).$$

If we allow ourselves to view the category $\operatorname{QCoh}(X)$ as a certain kind of ring, then more of our geometric objects appear affine. Namely, $\operatorname{QCoh}(X)^\heartsuit$ is classically a symmetric monoidal category, so it is basically a commutative ring object in the category of (small) categories. In the derived setting, we need to explain what sort of categories we want to output because it is dangerous to talk about the category of categories.

Definition 1.8. Let κ be a regular cardinal. Then an ∞-category $\mathcal{C}$ is *κ-presentable* if and only if every object is a small colimit of κ-compact objects, and $\mathcal{C}$ admits all colimits. We let Pr^L denote this category.

Written out, we let $\operatorname{Pr}^L_\kappa$ denote the ∞-category of presentable ∞-categories, and then

$$\operatorname{QCoh}\colon \operatorname{AffSch}^{\mathrm{op}} \to \operatorname{CAlg}(\operatorname{Pr}^L)$$

is fully faithful. (The functoriality implies that maps $X \to Y$ induces a map $f^*\colon \operatorname{QCoh}(Y) \to \operatorname{QCoh}(X)$ preserves $\otimes$ and colimits.) In fact, this functor continues to be fully faithful on larger categories.

Theorem 1.9. The fully faithful embedding $\operatorname{QCoh}\colon \operatorname{AffSch}^{\mathrm{op}} \to \operatorname{CAlg}(\operatorname{Pr}^L)$ extends to the category of qcqs schemes and more generally qcqs Artin stacks with affine diagonal.

Example 1.10. The sheaf $\mathrm{B}\mathbb{G}_m$ sending a ring R to its groupoid of invertible modules is a qcqs Artin stack with affine diagonal. (Being "Artin" comes from the fact that it is the quotient of a scheme by an algebraic action.) Note that $\mathrm{B}\mathbb{G}_m$ can be understood using the bar construction, which allows us to compute its category of quasicoherent sheaves. Namely, $\mathrm{B}\mathbb{G}_m = \operatorname{colim} \mathbb{G}_m^\bullet$, so

$$\operatorname{QCoh}(\mathrm{B}\mathbb{G}_m) = \lim \operatorname{QCoh}(\mathbb{G}_m^\bullet).$$

For example, modules on $\mathbb{G}_m^0$ are just abelian groups, and modules on $\mathbb{G}_m^1$ adds in some grading information. It turns out that $\operatorname{QCoh}(\mathrm{B}\mathbb{G}_m)$ is the category of graded abelian groups.

Notably, even though the ring of global sections cannot distinguish between $\mathrm{B}\mathbb{G}_m$ and $\operatorname{Spec}\mathbb{Z}$, the category of quasicoherent sheaves can tell the difference!

Remark 1.11. Let's check that a map $\operatorname{Spec} R \to \mathrm{B}\mathbb{G}_m$ has equivalent data to a map $\operatorname{QCoh}(\mathrm{B}\mathbb{G}_m) \to \operatorname{QCoh}(\operatorname{Spec} R)$. Well, $\operatorname{QCoh}(\mathrm{B}\mathbb{G}_m)$ is $\otimes$-generated by the graded abelian group $\mathbb{Z}[1]$, so the data of our map is uniquely determined by its image. Furthermore, $\mathbb{Z}[1]$ is $\otimes$-invertible, so it must go to an invertible sheaf on R. But the data of an invertible sheaf is exactly the required data of the map $\operatorname{Spec} R \to \mathrm{B}\mathbb{G}_m$!

Remark 1.12 (Tannakian duality)**.** Roughly speaking, QCoh being fully faithful on stacks of the form $\mathrm{B}G$ for algebraic groups G amounts to saying that the category

$$\mathrm{QCoh}(\mathrm{B}G) = \mathrm{Rep}\, G$$

determines the group G. This is Tannakian duality!

We now bravely assert that $\mathrm{QCoh}(X)$ is an algebraic object in itself; after all, it is a commutative ring object. Accordingly, we could decide that working with commutative rings was a mistake, and really we should have worked with $\mathrm{CAlg}(\mathrm{Pr}^L)$ from the start.

Remark 1.13 (Schlank)**.** The rings are dead. Long live the new rings.

1.1.3 Going Up

Our new category $\mathrm{CAlg}(\mathrm{Pr}^L)^{\mathrm{op}}$ is now so large that we could hope we don't have to do gluing anymore. After all, it already includes qcqs schemes.

It turns out that our new category is still not large enough for some applications.

Example 1.14. Note that $\mathrm{B}\mathbb{G}_m$ continues to be group object in stacks, so we can form $\mathrm{B}^2\mathbb{G}_m$. (Equivalently, we can just de-loop this group object.) The same descent procedure allows us to compute

$$\mathrm{QCoh}\left(\mathrm{B}^2\mathbb{G}_m\right) = \lim \mathrm{QCoh}\left(\mathrm{B}\mathbb{G}_m^\bullet\right).$$

However, all the extra descent structure is just describing a single grading structure on an abelian group, so the internal descent morphisms of the Čech nerve are all fully faithful, so $\mathrm{QCoh}\left(\mathrm{B}^2\mathbb{G}_m\right) = \mathrm{QCoh}(\mathrm{pt})$.

Remark 1.15. The stack $\mathrm{B}^2\mathbb{G}_m$ is not a pathological object: maps $X \to \mathrm{B}^2\mathbb{G}_m$ describe elements of $\mathrm{H}^2(X; \mathbb{G}_m)$.

Remark 1.16 (Schlank)**.** There are many reactions to this fact: denial, anger, bargaining, etc.

One could attempt to do geometry on $\mathrm{CAlg}(\mathrm{Pr}^L)^{\mathrm{op}}$ by doing some gluing, and it is not a bad idea.

Let's try to expand our algebraic category again.

Example 1.17. Let's examine $\mathrm{B}^2\mathbb{G}_m$ more seriously. A map $\mathrm{Spec}\, R \to \mathrm{B}^2\mathbb{G}_m$ amounts to the data of an Azumaya algebra A over R up to Morita equivalence, which means that we are really looking at the data of Mod_A. Now, we can think of Mod_A as a module over Mod_R, and being an Azumaya algebra amounts to asserting that

$$\mathrm{Mod}_{A^{\mathrm{op}}} \otimes_{\mathrm{Mod}_R} \mathrm{Mod}_A = \mathrm{Mod}_R.$$

Thus, a map $\mathrm{Spec}\, R \to \mathrm{B}^2\mathbb{G}_m$ seems to produce an invertible sheaf on Mod_R. The moral is that our algebraic category should be able to pick up on objects outputting $\mathrm{Mod}_{\mathrm{Mod}_R}(\mathrm{Pr}^L)$. A module over this ∞-category outputs an element of $\mathrm{CAlg}(2\mathrm{Pr}^L)$.

The moral is that we can enhance the $(\infty,1)$-categorical construction $R \mapsto \mathrm{Mod}_R$ to an $(\infty,2)$-categorical construction $R \mapsto \mathrm{Pr}^L_R$, where $\mathrm{Pr}^L_R := \mathrm{Mod}_{\mathrm{Mod}_R}(\mathrm{Pr}^L)$. For a general sheaf X on commutative rings, we define

$$\mathrm{Pr}^L_X := \lim_{\mathrm{Spec}\, R \to X} \mathrm{Pr}^L_R.$$

We constructed this bizarre object to deal with 2-categorical problems, so it is reasonable to hope that it can distinguish between $\mathrm{B}^2\mathbb{G}_m$ and the point.

Example 1.18. The same argument we used to compute $\mathrm{QCoh}(\mathrm{B}\mathbb{G}_m)$ via the Čech nerve can now compute $\mathrm{Pr}^L_{\mathrm{B}^2\mathbb{G}_m}$, and we find that it has equivalent data to the category of "graded (presentable) categories." Morally, what's going on is that we can do algebra in categories (with tensor products and so on), and $\mathrm{B}\mathbb{G}_m$ looks affine from the perspective of QCoh. However, we further discover that $\mathrm{Pr}^L_{\mathrm{B}^3\mathbb{G}_m} = \mathrm{Pr}^L_{\mathrm{pt}}$.

1.1.4 Stefanich Rings

We now see that we should not have stopped with $(\infty,1)$-categories, and we shouldn't stop with $(\infty,2)$-categories either.

Definition 1.19 (Stefanich ring)**.** A *Stefanich ring* is a tuple $(A_1, A_2, \ldots)$ such that $A_i \in \mathrm{CAlg}(i\mathrm{Pr}^L)$ for each i, and

$$A_n \cong \mathrm{End}_{A_{n+1}}(1).$$

The category of Stefanich rings is denoted StRing. Its opposite category is the category of *Gestalten*.

This definition does not quite make sense for size reasons, and we also need to explain what $i\mathrm{Pr}^L$ is. Let's discuss this.

Throughout the course, we fix a regular cardinal κ. In fact, we may as well take $\kappa = \aleph_1$.[1]

Remark 1.20. The category Pr^L_κ admits a product structure. It is equivalent to an object in $\mathrm{CAlg}(\mathrm{Pr}^L_\kappa)$. In short, one needs to know how to compute colimits of categories, and one needs to find a compact generator; it turns out that the arrow is a suitable generator.

The moral of the previous remark is that we can write down silly things like $\mathrm{Mod}_{\mathrm{Pr}^L_\kappa}(\mathrm{Pr}^L_\kappa)$ and take them seriously. Roughly speaking, this module category consists of presentable categories $\mathcal{C}$ equipped with an action map

$$\mathrm{Pr}^L \otimes \mathcal{C} \to \mathcal{C},$$

which is equivalent to the data of a functor $\mathrm{Pr}^L \times \mathcal{C} \to \mathcal{C}$ which is "bi-additive" in the sense that it preserves colimits in both coordinates. For example, one can choose presentable categories from Pr^L and receive functors $\mathcal{C} \to \mathcal{C}$. Alternatively, we can imagine choosing $X, Y \in \mathcal{C}$ and receiving a functor $(\mathrm{Pr}^L)^{\mathrm{op}} \to \mathrm{Sp}$ by $a \mapsto \mathrm{Hom}(a \otimes X, Y)$.

Notation 1.21. We define $2\mathrm{Pr}^L_\kappa := \mathrm{Mod}_{\mathrm{Pr}^L_\kappa}(\mathrm{Pr}^L_\kappa)$.

Note that Pr^L_κ is commutative, so it admits a tensor product, so $2\mathrm{Pr}^L_\kappa$ actually lives in $\mathrm{CAlg}(\mathrm{Pr}^L_\kappa)$. We can now inductively keep going.

The goal of the course is study these Stefanich rings. We have the following goals.

1. We want to explain that many old geometric objects can be found inside the opposite category of Stefanich rings. In other words, many old geometric objects are now affine.

2. No gluing is necessary: $\mathrm{StRing}^{\mathrm{op}}$ is essentially an ∞-topos, up to certain set-theoretic issues.

3. There is a notion of Cartier duality: given a Stefanich ring $(A_1, A_2, \ldots)$, one can restrict to $\otimes$-invertible objects, which is a sequence of spaces each of which is the loop space of the previous, so we get a spectrum!

[1] Taking $\kappa = \aleph_0$ is doable for some applications, but we want to do some analysis, so we want to deal with some limits of sequences and so on, so it is convenient to be able to access $\aleph_0$.

1.2 September 10

Today, we discuss how to do algebra with ∞-categories. Today's notes were copied (with gratitude) from Yutong Chen.

1.2.1 Presentable Categories

We are going to deal with size issues today, so let's recall a little set theory.

Definition 1.22 (regular)**.** Fix a cardinal κ. Then a set S is *κ-small* if and only if $|S| < \kappa$; in other words, there is an injection $S \hookrightarrow \kappa$. A cardinal κ is *regular* if and only if every κ-small union of κ-small sets is still κ-small.

Example 1.23. All successor cardinals of $\aleph_0$ are regular. For example, $\aleph_0$ is regular because a finite union of finite sets is finite. Similarly, $\aleph_1$ is regular because a countable union of countable sets is countable.

Remark 1.24. Let S be a set whose cardinality is a regular cardinal. Then the well-ordering principle allows us to order the elements of S as $\{s_\alpha\}_{\alpha\in\kappa}$, where κ has a natural ordering. This ordering produces a filtration

$$S = \bigcup_{\alpha\in\kappa} S_\alpha,$$

where each S_α is κ-small, and any κ-small subset T lives in S_α for some α. One can show this latter claim by a quick instance of transfinite induction on the size of T.

We now switch from sets to categories.

Definition 1.25 (filtered)**.** Fix a regular cardinal κ. A category I is *κ-filtered* if and only if every functor $D \to I$ from a κ-small category D factors through the category D^Δ, which is the category D together with a terminal object.

Remark 1.26. Roughly speaking, we are asking for the functor $F\colon D \to I$ to admit an object $y \in I$ such that each $x \in D$ has a morphism $Fx \to y$. (This is only a rough approximation because the maps $Fx \to y$ should also commute with each other.) For example, I is κ-filtered if I is closed under κ-small colimits.

Definition 1.27 (compact)**.** Fix a regular cardinal κ. Fix an ∞-category $\mathcal{C}$. An object $X \in \mathcal{C}$ is *κ-compact* if and only if $\mathrm{Hom}_{\mathcal{C}}(X, -)$ commutes with all κ-filtered colimits.

Remark 1.28. Intuitively, κ-compact means that the object is presented by fewer than κ elements.

Example 1.29. Let $\mathcal{C}$ be the (homotopy) category of abelian groups. Then a finitely presented abelian group is $\aleph_0$-compact.

Proposition 1.30. Fix a κ cardinal κ. If an ∞-cateogry is κ-compactly generated, then κ-filtered colimits commute with κ-small products.

Proof. Omitted. For example, we have not defined what it means to be compactly generated. ∎

Here is the main definition of this subsection.

Definition 1.31 (presentable)**.** Fix a regular cardinal κ. A category $\mathcal{C}$ is *κ-presentable* if and only if it has all κ-small colimits, and there is a set $S \subseteq \mathcal{C}$ of κ-compact objects for which each $X \in \mathcal{C}$ is a colimit of elements in S.

Remark 1.32. Any object in a κ-presentable category $\mathcal{C}$ can be written as a κ-filtered colimit of compact objects. Indeed, write $X = \operatorname{colim}_{i \in I} K_i$ where $K_i \in S$ for each i. Then the key is that $I = \operatorname{colim}_{\alpha \in \kappa} I_\alpha$ as in our set-theoretic discussion, so we are done upon writing

$$X = \operatorname*{colim}_{i \in I} K_i = \operatorname*{colim}_{\alpha \in \kappa} \operatorname*{colim}_{i \in I_\alpha} K_i.$$

One now checks that each object $\operatorname{colim}_{i \in I_\alpha} K_i$ is κ-compact.

1.2.2 A Category of Categories

Here are some basic structural properties.

Notation 1.33. Let $\mathcal{C}$ be a category, and let κ be a regular cardinal. Then we let $\mathcal{C}^\kappa$ denote the full subcategory of κ-compact objects.

Theorem 1.34. Let $\mathcal{C}$ be a κ-presentable category. Let $\operatorname{Ind}_\kappa \mathcal{C}^\kappa$ be the category obtained by formally adding κ-filtered colimits. Then the natural functor

$$\operatorname{Ind}_\kappa \mathcal{C}^\kappa \to \mathcal{C}$$

given by taking the formal colimits to the actual colimits is an equivalence.

Proof. We omit the proof. It is formal. ∎

Proposition 1.35. Let $\mathcal{C}$ be a κ-presentable category. Then $\mathcal{C}^\kappa$ is essentially κ-small.

Proof. For any κ-compact object $K \in \mathcal{C}$, we may write

$$K = \operatorname*{colim}_{i \in I} S_i$$

for some $S_i \in S$, where I is κ-filtered. By compactness, the natural map $K \to \operatorname{colim}_i S_i$ factors through S_i for some i, so K is a retract of S_i. Thus, we may enumerate the isomorphism classes in $\mathcal{C}^\kappa$ by enumerating the retracts of the objects in S, which is a κ-small collection. ∎

With a smallness condition, we can now build a category of categories.

Notation 1.36. Fix a regular cardinal κ. Let $\operatorname{Pr}^L_\kappa$ denote the category of κ-presentable categories, where morphisms are functors which preserve colimits and κ-compact objects. Furthermore, we define Pr^L to be the "union" of all $\operatorname{Pr}^L_\kappa$, which is the category of categories which are κ-presentable for some κ, where morphisms are functors which preserve colimits.

Remark 1.37. The union Pr^L has some size issues with its Hom-sets: for example,

$$\mathrm{Hom}_{\mathrm{Pr}^L}(\mathrm{Ani}, \mathcal{C}) = \mathcal{C}$$

for any $\mathcal{C} \in \mathrm{Pr}^L$. (Indeed, a functor out of Ani which preserves colimits is uniquely determined by where it sends the point.)

Notation 1.38. Fix a regular cardinal κ. Let Rex_κ be the category of all κ-small categories which admit κ-small colimits and retracts, where morphisms are functors preserving κ-small colimits.

Theorem 1.39 (Gabriel–Ulmer)**.** The natural functors $(-)^\kappa \colon \mathrm{Pr}^L_\kappa \to \mathrm{Rex}_\kappa$ and $\mathrm{Ind}_\kappa \colon \mathrm{Rex}_\kappa \to \mathrm{Pr}^L_\kappa$ are equivalences.

Sketch. We only discuss one of the inverse checks: we will show that $(\mathrm{Ind}_\kappa \mathcal{C})^\kappa = \mathcal{C}$. (The other inverse check was stated but not proved previously.) Well, start with some $\mathcal{C} \in \mathrm{Rex}_\kappa$. On one hand, one can check that $\mathcal{C} \subseteq \mathrm{Ind}_\kappa \mathcal{C}$ maps to the κ-compact objects, due to the formal nature of the colimits we added. On the other hand, for any object $X \in \mathrm{Ind}_\kappa \mathcal{C}$ which is κ-compact, we may write $X = \mathrm{colim}_{\alpha \in I} D_\alpha$ for some κ-filtered I by construction of $\mathrm{Ind}_\kappa \mathcal{C}$; then X is the retract of one of the objects D_α, so $X \in \mathcal{C}$ by definition of Rex_κ. ■

Remark 1.40. Theorem 1.39 tells us that the Hom-sets in Pr^L_κ do not have the size issues that Pr^L does: indeed,

$$\mathrm{Hom}_{\mathrm{Pr}^L_\kappa}(\mathcal{C}, \mathcal{D}) = \mathrm{Hom}_{\mathrm{Rex}_\kappa}(\mathcal{C}^\kappa, \mathcal{D}^\kappa),$$

and the target is κ-small. For example, each of $\mathcal{C}^\kappa$ and $\mathcal{D}^\kappa$ are essentially κ-small by Proposition 1.35.

We will view Pr^L_κ as our "category of categories." For example, one desirable feature of this definition is that it is (essentially) an element of itself.

Theorem 1.41. The category Pr^L_κ is κ-presentable. We may write this mathematical statement as $\mathrm{Pr}^L_\kappa \in \mathrm{Pr}^L_\kappa$.

Sketch. By Theorem 1.39, it is enough to show that Rex_κ is κ-presentable. We will study the forgetful functor $U \colon \mathrm{Rex}_\kappa \to \mathrm{Cat}$. For example, it reflects isomorphisms. Also, Theorem 1.39 implies that $F \colon \mathrm{Cat}_\infty \to \mathrm{Rex}_\kappa$ defined by

$$F(S) \coloneqq \mathrm{PSh}(S; \mathrm{Ani})^\kappa$$

is a left adjoint for F. Here, F is better thought of as the corresponding functor $\mathrm{PSh} \colon \mathrm{Cat}_\infty \to \mathrm{Pr}^L_\kappa$. I don't want to think about the adjunction checks.

Now, Cat_∞ is generated by the κ-compact object Δ^n. Also, the functor F sends κ-compact objects to κ-compact objects. Thus, $\{F\Delta^n\}_n$ are κ-compact objects, and we would like them to generate. To this end, define $T \coloneqq FG$, so there is a natural map $T \to 1$ given by the adjunction. It turns out that the natural map

$$T^2\mathcal{C} \to T\mathcal{C}$$

has co-equalizer $\mathcal{C}$, which completes the proof by writing the relevant objects in Cat_∞ as colimits of Δ^ns. ■

1.2.3 Algebra with Higher Categories

Another reason why presentability is a desirable hypothesis is the presence of an adjoint functor theorem.

Theorem 1.42 (Lurie)**.** Let $F : \mathcal{C} \to \mathcal{D}$ be a morphism in Pr^L_κ.

(a) Then F admits a right adjoint if and only if it commutes with small colimits.

(b) Then F admits a left adjoint if and only if F commutes with small limits and is accessible, meaning that F commutes with κ-filtered colimits for some regular cardinal κ.

Remark 1.43. Roughly speaking, there is an asymmetry in (a) and (b) because presentability is about generation by colimits. If one instead defined the corresponding category Pr^R_κ (whose morphisms preserve accessible functors), then we would expect the asymmetry to happen in the other direction.

Remark 1.44. It turns out that $\mathrm{Pr}^R_\kappa \cong \mathrm{Pr}^R$, and the forgetful functor $\mathrm{Pr}^R \to \mathrm{Cat}_\infty$ preserves small limits. One can use this to show that Pr^L admits small colimits.

Example 1.45. Coproducts in Pr^L are (naïve) products. Products are also naïve products.

Example 1.46. Products in Pr^L_κ are Ind-saturations of naïve products.

Products and coproducts being the same means that we will have to work harder to build a tensor product.

Definition 1.47 (Lurie tensor product)**.** Fix $\mathcal{C}, \mathcal{D} \in \mathrm{Pr}^L_\kappa$. Then we define $\mathcal{C} \otimes \mathcal{D}$ by universal property: we have

$$\mathrm{Hom}_{\mathrm{Pr}^L_\kappa}(\mathcal{C} \otimes \mathcal{D}, -) \cong \mathrm{Bilin}_{\mathrm{Pr}^L_\kappa}(\mathcal{C} \times \mathcal{D}, -).$$

Here, "Bilin" refers to the functors linear in both coordinates.

Remark 1.48. One can construct $\mathcal{C} \otimes \mathcal{D}$ via the adjoint functor theorem. Namely, the functor $\mathcal{C} \otimes -$ is an adjoint to the corresponding bilinear functor.

1.3 September 17

Today, we discuss dualizable presentable ∞-categories.

1.3.1 Tensor Products

Let's return to tensor products. We are hunting for a symmetric monoidal structure on Pr^L_κ.

Theorem 1.49. There is a unique functor

$$- \otimes - : \mathrm{Pr}^L \times \mathrm{Pr}^L \to \mathrm{Pr}^L$$

such that

$$\mathrm{Hom}_{\mathrm{Pr}^L}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) = \mathrm{Bilin}_{\mathrm{Pr}^L}(\mathcal{C} \times \mathcal{D}, \mathcal{E}).$$

Here, Bilin means that we have functors $\mathcal{C} \times \mathcal{D} \to \mathcal{E}$ which preserve colimits in each variable.

Proof. We will directly construct $\otimes$. For any $\mathcal{C} \in \mathrm{Pr}^L$, we may choose a presentation $\mathcal{C} = \langle S \rangle \left[R^{-1}\right]$, where S is a small ∞-category, and R is a collection of morphisms. (Indeed, such a presentation exists by writing out how $\mathcal{C}$ is compactly generated. For example, $\langle S \rangle$ is the "free presentable" category on S, which is $\mathrm{PSh}(S; \mathrm{Ani})$.) Accordingly, given $\mathcal{C}$ and $\mathcal{D}$ with presentations $\mathcal{C} = \langle S_{\mathcal{C}} \rangle \left[R_{\mathcal{C}}^{-1}\right]$ and $\mathcal{D} = \langle S_{\mathcal{D}} \rangle \left[R_{\mathcal{D}}^{-1}\right]$, we define

$$\mathcal{C} \otimes \mathcal{D} \coloneqq \langle S_{\mathcal{C}} \times S_{\mathcal{D}} \rangle \left[(R_{\mathcal{C}} \times 1 \cup 1 \times R_{\mathcal{D}})^{-1}\right].$$

One can check the universal property from here fairly directly. ■

Remark 1.50. There is some intuition that colimits should give our additive structure, which motivates our definition of Bilin.

Remark 1.51. In fact, we have the adjunction property

$$\mathrm{Hom}_{\mathrm{Pr}^L}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) = \mathrm{Hom}_{\mathrm{Pr}^L}(\mathcal{C}, \mathrm{Fun}(\mathcal{D}, \mathcal{E})).$$

Indeed, it is enough to check that the right-hand set is $\mathrm{Bilin}_{\mathrm{Pr}^L}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$, which we see by currying.

Remark 1.52. If $\mathcal{C}$ and $\mathcal{D}$ are κ-presentable, then $\mathcal{C} \otimes \mathcal{D}$ is also κ-presentable. Indeed, being κ-presentable allows us to write $\mathcal{C} = \langle \mathcal{C}^{\kappa} \rangle \left[R_{\mathcal{C}}^{-1}\right]$ and $\mathcal{D} = \langle \mathcal{D}^{\kappa} \rangle \left[R_{\mathcal{D}}^{-1}\right]$, so our explicit construction of $\mathcal{C} \otimes \mathcal{D}$ tells us that it is generated by κ-compact objects, given by inverting only morphisms between κ-compact objects. In fact, one can check that $- \otimes -$ restricts to a functor $\mathrm{Pr}^L_{\kappa} \times \mathrm{Pr}^L_{\kappa} \to \mathrm{Pr}^L_{\kappa}$: one only needs to check that it sends compact objects to compact objects, which can be done via the presentation.

Example 1.53. Suppose $\mathcal{C} = \langle S \rangle$ so that $\mathcal{C} = \mathrm{PSh}(S, \mathrm{Ani})$. Then for any $\mathcal{D} \in \mathrm{Pr}^L$, we have

$$\mathcal{C} \otimes \mathcal{D} \cong \mathrm{PSh}(S, \mathcal{D}).$$

For example, this can be seen directly from the presentation.

The example has supplied us with "dualizable" objects.

Definition 1.54 (dualizable)**.** Let X be an object of a symmetric monoidal category. Then X is *dualizable* if and only if there is an object $X^{\vee}$ such that $X^{\vee} \otimes - \cong \mathrm{Mor}(X, -)$.

Remark 1.55. It is equivalent to ask that $X \otimes -$ admits a left adjoint of the form $X^{\vee} \otimes -$. One sees that the definitions are equivalent by moving around adjoints and using the $\otimes$-Hom adjunction.

Remark 1.56. By unwinding the adjunctions, we are provided with a co-evaluation $1 \to X \otimes X^{\vee}$ and an evaluation $X^{\vee} \otimes X \to 1$ for which the composite

$$X \to X \otimes X^{\vee} \otimes X \to X$$

is the identity. This data is basically equivalent to asserting that $X^{\vee}$ is the dual of X.

Example 1.57. Example 1.53 implies that $\mathrm{PSh}(S; \mathrm{Ani})$ is dualizable for any small S.

1.3.2 Dualizable Stable Presentable Categories

As motivation, let's work with some stable categories.

Definition 1.58 (stable)**.** Define $\mathrm{Pr}^L_{\mathrm{st}}$ to be the category of *stable presentable* categories, meaning that the objects are presentable categories which admit finite limits, finite colimits, a zero object, and all pushout squares are also pullback squares. Morphisms in $\mathrm{Pr}^L_{\mathrm{st}}$ are required to be exact and preserve colimits.

Remark 1.59. The internal hom in $\mathrm{Pr}^L_{\mathrm{st}}$ turns out to agree with the internal hom in Pr^L.

Remark 1.60. If S is a small stable category, then the free presentable stable ∞-category is $\mathrm{Ind}\, S$; this turns out to be the same as the category of exact presheaves on S (valued in Sp).

Proposition 1.61. Suppose $\mathcal{C} \in \mathrm{Pr}^L_{\mathrm{st}}$ is κ-compactly generated. Then $\mathcal{C}$ is dualizable with dual $\mathrm{Ind}(\mathcal{C}^{\kappa,\mathrm{op}})$.

Proof. Define $\mathcal{C}^\vee \coloneqq \mathrm{Ind}(\mathcal{C}^{\kappa,\mathrm{op}})$ for brevity. Then one computes that

$$\mathcal{C}^\vee \otimes D = \mathrm{Fun}^{\mathrm{ex}}(\mathcal{C}^\kappa, \mathcal{D})$$

because $\mathcal{C}^\vee$ is the free stable category on $\mathcal{C}^\kappa$. But now the right-hand side is simply $\mathrm{Fun}^L(\mathrm{Ind}(\mathcal{C}^\kappa), \mathcal{D})$, which is $\mathrm{Fun}^L(\mathcal{C}, \mathcal{D})$. Thus, we are done! ∎

Our goal is to classify dualizable presentable categories. For example, it will include the stable ones.

Definition 1.62. A map $f\colon X \to Y$ of categories is *compact* if and only if the following holds: for all filtered limits $Z = \mathrm{colim}_i Z_i$, there is a dashed arrow making the diagram

$$\begin{array}{ccc} \operatorname*{colim}_i \mathrm{Hom}(Y, Z_i) & \longrightarrow & \mathrm{Hom}(Y, Z) \\ \downarrow{\scriptstyle f} & \swarrow & \downarrow{\scriptstyle f} \\ \operatorname*{colim}_i \mathrm{Hom}(X, Z_i) & \longrightarrow & \mathrm{Hom}(X, Z) \end{array}$$

commute.

Example 1.63. One can check that X is compact if and only if $\mathrm{id}_X\colon X \to X$ is compact.

Example 1.64. If f is a compact morphism, then any composite with f is still compact.

Example 1.65. A morphism of vector spaces is compact if and only if the image is finite-dimensional.

Lemma 1.66. Suppose that $\mathcal{C} \in \mathrm{Pr}^L_\omega$. Then a map $f\colon X \to Y$ in $\mathcal{C}$ is compact if and only if f factors through a compact object.

Proof. The backward direction follows from the two examples. For the forward direction, suppose f is compact. We may write Y as a filtered colimit $\mathrm{colim}_i Z_i$, where each Z_i is compact. Then compactness of f implies that f factors through Z_i for some i. ∎

Definition 1.67 (compactly assembled)**.** An object X of a presentable category $\mathcal{C}$ is *basic nuclear* if and only if X is a colimit of a diagram of the form

$$X_0 \to X_1 \to X_2 \to \cdots,$$

where each morphism $X_i \to X_{i+1}$ is compact. The category $\mathcal{C}$ is *compactly assembled* if and only if it is generated (by colimits) by basic nuclear objects.

Example 1.68. Let $\mathcal{C}$ be the poset category of open subsets of $\mathbb{R}$. By Lemma 1.66, $U \subseteq V$ is compact if and only if there is a compact set between them. It turns out that $\mathcal{C}$ is compactly assembled.

Proposition 1.69. Basic nuclear objects of a presentable category are $\aleph_1$-compact.

Proof. Write a basic nuclear object X as promised as $\operatorname{colim}_i X_i$. Now, suppose that $Z = \operatorname{colim}_n Z_n$ is an $\aleph_1$-filtered colimit. We now compute

$$\begin{aligned}\operatorname{Hom}(X, Z) &= \operatorname{Hom}\left(\operatorname*{colim}_i X_i, \operatorname*{colim}_n Z_n\right) \\ &= \lim_i \operatorname{Hom}(X_i, \operatorname*{colim}_n Z_n) \\ &= \lim_i \operatorname*{colim}_n \operatorname{Hom}(X_i, Z_n).\end{aligned}$$

Now, one can swap the limit and colimit here using the maps

$$\operatorname{Hom}(X_i, \operatorname{colim} Z_n) \to \operatorname{colim} \operatorname{Hom}(X_i, Z_n)$$

given by compactness. We conclude that

$$\operatorname{Hom}(X, Z) = \operatorname*{colim}_n \operatorname{Hom}(X, Z_n),$$

as desired. ∎

Remark 1.70. If $\mathcal{C}$ is compactly generated, then the converse is also true.

Here is our main theorem.

Theorem 1.71. Let $\mathcal{C}$ be a presentable stable category. Then the following are equivalent.

(i) $\mathcal{C}$ is dualizable.

(ii) $\mathcal{C}$ is retract of a compactly generated category.

(iii) $\mathcal{C}$ is $\aleph_1$-presentable, and the natural map $\operatorname{colim}\colon \operatorname{Ind}(\mathcal{C}^{\aleph_1}) \to \mathcal{C}$ admits a left adjoint.

(iv) $\mathcal{C}$ is compactly assembled.

Sketch. Here we go.

- For (iii) implies (ii), being a fully faithful right adjoint means that the map is localization, so admitting a left adjoint induces a splitting from the localization.
- For (ii) implies (i), note that compactly generated categories are dualizable, and dualizability can be checked to be stable under taking retracts.

- For (i) implies (iii), again use the fact that $\mathrm{Ind}(\mathcal{C}^{\aleph_1}) \to \mathcal{C}$ is a localization. Thus,
$$\mathcal{C}^\vee \otimes \mathrm{Ind}(\mathcal{C}^{\aleph_1}) \to \mathcal{C}^\vee \otimes \mathcal{C}$$
is also a localization. This is equivalent to
$$\mathrm{Fun}^L\left(\mathcal{C}, \mathrm{Ind}(\mathcal{C}^{\aleph_1})\right) \to \mathrm{Fun}^L(\mathcal{C}, \mathcal{C})$$
being a localization. Thus, we may lift morphisms; lifting $\mathrm{id}_{\mathcal{C}}$ produces the needed adjunction.
- For (iv) implies (iii), we need to define a map $\mathcal{C} \to \mathrm{Ind}\left(\mathcal{C}^{\aleph_1}\right)$. It is enough to define the map on basic nuclear objects, for which send $X = \mathrm{colim}_i X_i$ to the corresponding colimit in $\mathrm{Ind}\left(\mathcal{C}^{\aleph_1}\right)$. The main check is that
$$\mathrm{Hom}_{\mathcal{C}}\left(\operatorname*{colim}_i X_i, \operatorname*{colim}_n Z_n\right) = \mathrm{Hom}_{\mathrm{Ind}(\mathcal{C}^{\aleph_1})}\left(\operatorname*{colim}_i X_i, \operatorname*{colim}_n Z_n\right),$$
which is purely formal.
- For (iii) implies (iv), reverse the argument of the previous point to write any $X \in \mathcal{C}^{\aleph_1}$ to some colimit of the form $\mathrm{colim}\, X_i$ in $\mathrm{Ind}\left(\mathcal{C}^{\aleph_1}\right)$. One finds that the maps $X_i \to X$ are compact, which is enough. ■

Now that we understand dualizable objects, we feel comfortable making them a category.

Definition 1.72. The category $\left(\mathrm{Pr}^L_{\mathrm{st}}\right)^{\mathrm{dual}}$ is the subcategory of dualizable $\mathcal{C} \in \mathrm{Pr}^L_{\mathrm{st}}$, where morphisms are functors $\mathcal{C} \to \mathcal{D}$ whose right adjoints admit right adjoints.

Remark 1.73. If $\mathcal{C}$ and $\mathcal{D}$ are compactly generated, then the presence of all these right adjoints means that compact objects are sent to compact objects.

Remark 1.74. It turns out that there is a functorial splitting of $\mathcal{C} \to \mathcal{C}$ (for dualizable $\mathcal{C}$) through $\mathrm{Ind}\left(\mathcal{C}^{\aleph_1}\right)$.

Proposition 1.75. The forgetful functor to $\left(\mathrm{Pr}^L_{\mathrm{st}}\right)^{\mathrm{dual}} \to \mathrm{Pr}^L_{\mathrm{st}}$ commutes with colimits.

Proof. Use the functorial splitting of Remark 1.74. The moral is that we should check that colimits of dualizable categories continue to be dualizable. Well, any such colimit $\mathrm{colim}_n \mathcal{C}_n$ admits a factorization
$$\operatorname*{colim}_n \mathcal{C}_n \to \operatorname*{colim}_n \mathrm{Ind}\, \mathcal{C}_n^{\aleph_1} \to \operatorname*{colim}_n \mathcal{C}_n,$$
but the middle term is $\mathrm{Ind}\,\mathrm{colim}_n \mathcal{C}_n^{\aleph_1}$. In particular, the middle term can be seen to be dualizable, and Theorem 1.71 tells us that the retract $\mathrm{colim}_n \mathcal{C}_n$ is still compactly assembled. ■

Here are some more facts, whose proofs we omit.

Proposition 1.76. The category $\left(\mathrm{Pr}^L_{\mathrm{st}}\right)^{\mathrm{dual}}$ is $\aleph_1$-presentable.

Proposition 1.77. Fix an uncountable regular cardinal κ and some dualizable stable presentable $\mathcal{C}$. Then the following are equivalent.

(i) $\mathcal{C}$ is κ-compact in $\left(\mathrm{Pr}^L_{\mathrm{st}}\right)^{\mathrm{dual}}$.

(ii) $\mathcal{C}$ is κ-compact in $\mathrm{Pr}^L_{\mathrm{st},\kappa}$.

(iii) $\mathcal{C}$ is dualizable in $\mathrm{Pr}^L_{\mathrm{st},\kappa}$.

1.3.3 Dualizable $\mathcal{V}$-Linear Categories

Throughout, we fix some commutative algebra object $\mathcal{V} \in \mathrm{Pr}^L_{\aleph_1}$.

Notation 1.78. Define $\mathrm{Pr}^L_{\mathcal{V}} \coloneqq \mathrm{Mod}_{\mathcal{V}}(\mathrm{Pr}^L)$.

Definition 1.79 (internal left adjoint)**.** A map $f\colon M \to N$ in $\mathrm{Pr}^L_{\mathcal{V}}$ is an *internal left adjoint* if and only if its right adjoint f^R commutes with colimits, and the induced map

$$v \otimes f^R(x) \to f^R(v \otimes x)$$

is an equivalence for all $v \in \mathcal{V}$ and $x \in N$.

Definition 1.80 (atomic)**.** Fix $\mathcal{M} \in \mathrm{Pr}^L_{\mathcal{V}}$. Then $X \in \mathcal{M}$ is *$\mathcal{V}$-atomic* if and only if $x \otimes -\colon \mathcal{V} \to \mathcal{M}$ is an internal left adjoint. Let $\mathcal{M}^{\mathrm{at}}$ denote the full subcategory of atomic objects.

Example 1.81. By expanding out the definition, we see that if $\mathcal{V} = D(\mathbb{S})$, then $\mathcal{V}$-atomic is equivalent to compact.

Example 1.82. If $X \in \mathcal{V}$, then X is atomic

Thus, we hope that we can replace the word "compact" with the word "atomic" everywhere in the previous subsection and prove some nice theorems.

Definition 1.83 (atomically generated)**.** A category $\mathcal{M} \in \mathrm{Pr}^L_{\mathcal{V}}$ is *atomically generated* if and only if $\mathcal{M}$ is generated under colimits (and tensor products with $\mathcal{V}$) by atomic objects.

Definition 1.84. For any small $\mathcal{V}$-enriched category S, define $P_{\mathcal{V}}(S)$ to be the free presentable $\mathcal{V}$-module on S.

Remark 1.85. If $\mathcal{M}$ is $\mathcal{V}$-atomically generated, then it turns out that $\mathcal{M} \cong P_{\mathcal{V}}(\mathcal{M}^{\mathrm{at}})$.

Proposition 1.86. If $\mathcal{M}$ is $\mathcal{V}$-atomically generated, then $\mathcal{M}$ is dualizable, and its dual is $P_V(\mathcal{M}^{\mathrm{at,op}})$.

Proof. We rewrite the proof of Proposition 1.61: indeed,

$$\begin{aligned}
\mathcal{M}^\vee \otimes \mathcal{N} &\cong P_{\mathcal{V}}\left(\mathcal{M}^{\mathrm{at,op}}\right) \otimes \mathcal{N} \\
&\cong \mathrm{Fun}_{\mathcal{V}}\left(\mathcal{M}^{\mathrm{at}}, \mathcal{N}\right) \\
&\cong \mathrm{Fun}^L_{\mathcal{V}}\left(P_V(\mathcal{M}^{\mathrm{at}}), \mathcal{N}\right) \\
&\cong \mathrm{Fun}^L_{\mathcal{V}}(\mathcal{M}, \mathcal{N}),
\end{aligned}$$

as desired. ∎

Theorem 1.87. Fix $\mathcal{M} \in \mathrm{Pr}^L_{\mathcal{V}}$. Then the following are equivalent.

(i) $\mathcal{M}$ is dualizable.

(ii) $\mathcal{M}$ is a retract of a $\mathcal{V}$-atomically generated category $\mathcal{N} \in \mathrm{Pr}^L_{\mathcal{V}}$.

(iii) $\mathcal{M}$ is $\aleph_1$-compactly generated, and the natural map $P_V(\mathcal{M}^{\aleph_1}) \to \mathcal{M}$ admits a left adjoint.

Proof. Rewrite the proof of Theorem 1.71; note that we are avoiding part (iv). I didn't understand the proof the first time, so I will not bother to rewrite it. ■

Remark 1.88. For (iii), the proof shows that the left adjoint is internal.

Definition 1.89. Let $\left(\mathrm{Pr}^L_{\mathcal{V}}\right)^{\mathrm{dual}}$ be the subcategory of $\mathrm{Pr}^L_{\mathcal{V}}$ whose objects are dualizable and whose morphisms are $\mathcal{V}$-linear internal left adjoints.

Remark 1.90. It turns out that $\mathcal{V}$-linear internal left adjoints preserves compact objects.

Remark 1.91. One has the same functorial splitting

$$\mathcal{M} \to P_{\mathcal{V}}\left(\mathcal{M}^{\aleph_1}\right) \to \mathcal{M}$$

in $\mathrm{Pr}^L_{\mathcal{V}}$ as before.

One has the same facts as before. Here is one of them.

Proposition 1.92. The forgetful functor $\left(\mathrm{Pr}^L_{\mathcal{V}}\right)^{\mathrm{dual}} \to \mathrm{Pr}^L_{\mathcal{V}}$ commutes with colimits, and $\left(\mathrm{Pr}^L_{\mathcal{V}}\right)^{\mathrm{dual}}$ is $\aleph_1$-presentable.

1.4 October 1

This talk was given by Jeremey Hahn. We will discuss quasicoherent sheaves.

1.4.1 Stefanich Rings

As usual, we denote by Pr^L_{κ} the category of κ-presentable categories, κ is some regular cardinal. We will mostly take $\kappa := \aleph_1$. The morphisms are given by left adjoints which preserve κ-compact objects.

Remark 1.93. Professor Hahn's $\aleph$s are immaculate.

Remark 1.94. It is convenient to use $\aleph_1$ instead of (say) $\aleph_0$ because $\aleph_1$-filtered colimits commute with countable limits, and countable limits appear frequently. For example, one may want to take the totalization of a cosimplicial object, which is a countable limit.

Example 1.95. Let $\mathcal{C}$ be the homotopy category of derived p-complete abelian groups. (In short, $\mathcal{C}$ is the category of abelian groups which are isomorphic to their own p-completions.) For example, $\mathbb{Z}_p$ is a projective generator of $\mathcal{C}$. However, $\mathbb{Z}_p$ is not ω-compact: taking $\mathrm{Hom}(\mathbb{Z}_p, -)$ on the diagram

$$\operatorname{colim}\left(\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \cdots\right).$$

On one hand, $\mathrm{Hom}(\mathbb{Z}_p, \operatorname{colim} \mathbb{Z}_p) = \mathrm{Hom}(\mathbb{Z}_p, \mathbb{Q}_p^\wedge) = 0$. On the other hand, $\operatorname{colim} \mathrm{Hom}(\mathbb{Z}_p, \mathbb{Z}_p) = \mathbb{Q}_p$.

Recall that Pr^L_κ admits a tensor product. We will be interested in commutative algebra objects $\mathcal{C}$ in Pr^L_κ, which admit module categories $\mathrm{Mod}_{\mathcal{C}} \mathrm{Pr}^L_\kappa$.

Remark 1.96. Roughly speaking, a category in $\mathrm{Mod}_{\mathcal{C}} \mathrm{Pr}^L_\kappa$ is enriched over $\mathcal{C}$.

In general, module categories acquire a symmetric monoidal structure. In other words, $\mathrm{Mod}_{\mathcal{C}} \mathrm{Pr}^L_\kappa$ is a commutative algebra object in Pr^L_κ.

Theorem 1.97. There is a fully faithful map $\mathrm{CAlg}(\mathcal{C}) \to \mathrm{CAlg}(\mathrm{Mod}_{\mathcal{C}} \mathrm{Pr}^L_\kappa)$ (in Pr^L_κ) given by sending A to $\mathrm{Mod}_A \mathcal{C}$.

Remark 1.98. To be a morphism, this map needs a right adjoint. The right adjoint sends $\mathcal{D}$ to $\mathrm{Hom}_{\mathcal{D}}(1)$. Because $\mathcal{D}$ is $\mathcal{C}$-linear, $\mathrm{Hom}_D(1)$ upgrades to a $\mathcal{C}$-linear commutative algebra object.

We are now ready to define a tower.

Definition 1.99. Define $1\,\mathrm{Pr} \coloneqq \mathrm{Pr}^L_{\aleph_1}$. Then for each $n \geq 1$, define $(n+1)\,\mathrm{Pr} \coloneqq \mathrm{Mod}_{n\mathrm{Pr}}(1\mathrm{Pr})$. Intuitively, these are the $n\mathrm{Pr}$-linear presentable categories.

Remark 1.100. There are fully faithful functors

$$\mathrm{CAlg}(1\mathrm{Pr}) \to \mathrm{CAlg}(2\mathrm{Pr}) \to \cdots$$

provided by Theorem 1.97.

This lets us cleanly define Stefanich rings.

Definition 1.101. Define the category of *Stefanich rings* to be the $\aleph_1$-presentable symmetric monoidal category

$$\operatorname{colim}\left(\mathrm{CAlg}(1\mathrm{Pr}) \to \mathrm{CAlg}(2\mathrm{Pr}) \to \cdots\right),$$

where the colimit takes place in $1\mathrm{Pr}$.

Remark 1.102. Colimits in $1\mathrm{Pr}$ have underlying ∞-category given by the limit in just the category of ∞-categories (of the right adjoints). It follows that Stefanich rings can be realized as sequences $(A_1, A_2, \ldots)$ where $A_n \in \mathrm{CAlg}(n\mathrm{Pr})$, and we have chosen isomorphisms $A_{n-1} \cong \mathrm{End}_{A_n}(1)$. (This latter isomorphism upgrades Remark 1.98.)

1.4.2 Fun with $\mathbb{E}_\infty$-Rings

Here is an A-linear variant.

Example 1.103. Suppose A is an $\mathbb{E}_\infty$-ring. Then $\mathrm{Mod}_A(\mathrm{Spaces}) = D(A)$, which is a commutativ algebra object in $1\mathrm{Pr}$. Accordingly, we define $1\mathrm{Pr}_A \coloneqq \mathrm{Mod}_{D(A)}(1\mathrm{Pr})$ and then continue inductively by $(n+1)\mathrm{Pr}_A \coloneqq \mathrm{Mod}_{n\mathrm{Pr}}(1\mathrm{Pr})$.

Remark 1.104. Note that $n\mathrm{Pr}_A = \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr})$. There is a natural map of commutative algebra objects $(n-1)\mathrm{Pr} \to (n-1)\mathrm{Pr}_A$, so we can nest our modules to make this category

$$\mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(\mathrm{Mor}_{(n-1)\mathrm{Pr}}(1\mathrm{Pr})) = \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(n\mathrm{Pr}).$$

This is now a commutative algebra object in $n\mathrm{Pr}$. Thus, $\{n\mathrm{Pr}_A\}_n$ is a Stefanich ring.

Remark 1.105. For each $n \geq 0$, the map sending an $\mathbb{E}_\infty$-ring A to $n\mathrm{Pr}_A \in \mathrm{CAlg}(1\mathrm{Pr})$ upgrades to a functor.

Remark 1.106. It turns out that $n\mathrm{Pr}_A$ has a $\otimes$-Hom adjunction. Also, for a map $f\colon A \to B$ of $\mathbb{E}_\infty$-rings, there is a pullback functor $f_n^*\colon n\mathrm{Pr}_A \to n\mathrm{Pr}_B$. (This is basically a tensor product. We already met this map in Remark 1.104.) There is also a functor f_{n*}, which is basically restriction.

Here are some more difficult elaborations on this remark. All of these results are proved by induction on n. Typically, the base case is harder than the induction.

Proposition 1.107. Let $f\colon A \to B$ be a map of $\mathbb{E}_\infty$-rings. Then $f_n^*\colon n\mathrm{Pr}_A \to n\mathrm{Pr}_B$ admits an $n\mathrm{Pr}_A$-linear colimit-preserving right adjoint f_{n*}. Moreover, f_{n*} commutes with base-change and satisfies a projection formula.

Proof. Let's argue this for $n = 0$. Then we are on the hunt for a functor $f_*\colon D(B) \to D(A)$, which we construct as restriction of scalars along f. This provides the adjoint, so it only remains to check that f_* preserves colimits; this holds because B-modules can already be computed in $\mathrm{Spectra}$. ∎

Definition 1.108. A category $\mathcal{C}$ is *countably presented* if and only if it is $\aleph_1$-compactly generated.

Proposition 1.109. If B is a countably presented A-algebra, then f_{n*} preserves $\aleph_1$-compact objects.

Proof. Let's start with $n = 0$. Let $D(B)^\kappa$ denote the set of compact objects. This turns out to be the subcategory generated by B under countable colimits and shifts. Thus, f_{0*} preserving compact objects is equivalent to checking that B is κ-compact as an A-module. Well, because B is countably presented, we can realize B as a countable colimit of algebras of the form

$$\mathrm{Free}_A(M) = \bigoplus_{n\in\mathbb{N}} M^{\otimes_A n}_{h\Sigma_n},$$

where M is a compact A-module.

In order to have done this at least once, we indicate how to do the induction on n. By the induction, it turns out that we may suppose $(n-1)\mathrm{Pr}_A \to (n-1)\mathrm{Pr}_B$ is countably presented. Now, we must show that the map $f_{n*}\colon n\mathrm{Pr}_B \to n\mathrm{Pr}_A$ sends compact objects to compact objects. It turns out that the compact objects are generated by objects of the form $X \otimes (n-1)\mathrm{Pr}_B$ where X is compact in $n\mathrm{Pr}$, so it basically suffices to handle $n\mathrm{Pr}_B$. Then we apply an argument similar to the one of the previous paragraph. ∎

Remark 1.110. In the setting of Proposition 1.109, we may write $f_n^!$ for the additional right adjoint of f_{n*}.

Lemma 1.111. The functor $A \mapsto n\mathrm{Pr}_A$ commutes with finite limits.

Proof. At $n = 0$, this is basically asking us to show that $D(A \times B) = D(A) \times D(B)$. This is true because $A \times B$ admits idempotents which allows us to recover any module as a product of an A-module and a B-module.

We will not do the whole induction, but let's indicate $n = 1$. This amounts to checking

$$\mathrm{Mod}_{R\times S}(1\mathrm{Pr}) = \mathrm{Mod}_R(1\mathrm{Pr}) \times \mathrm{Mod}_S(1\mathrm{Pr}).$$

Thus, we are trying to say that a category which is $(R \times S)$-linear admits linearity by R and S separately. The moral is that R and S themselves are idempotent objects, so we can take tensor products to exhibit these equivalences. ■

Lemma 1.112. Let $f\colon A \to B$ be a countably presented map of $\mathbb{E}_\infty$-algebras. For $n \geq 1$, the object $f_{n*}(1) \in n\mathrm{Pr}_A$ is dualizable.

Proof. This is an application of Morita theory. Note $f_{n*}(1) = (n-1)\mathrm{Pr}_B$. In fact, this object is self-dual. The evaluation map needs to map out of $(n-1)\mathrm{Pr}_B \otimes (n-1)\mathrm{Pr}_B$ (where the tensor product takes place in $n\mathrm{Pr}_A$). Taking modules preserves colimits, so this is just $(n-1)\,\mathrm{Pr}_{B\otimes_A B}$; there is then a diagonal map to $(n-1)\mathrm{Pr}_B$, which we then push forward to $(n-1)\mathrm{Pr}_A$.

Analogously, the co-evaluation functor is given by

$$(n-1)\mathrm{Pr}_A \xrightarrow{f^*} (n-1)\mathrm{Pr}_B \xrightarrow{\Delta_*} (n-1)\mathrm{Pr}_{B\otimes_A B} \cong (n-1)\mathrm{Pr}_B \otimes_{(n-1)\mathrm{Pr}_A} (n-1)\mathrm{Pr}_B.$$

One then checks that these evaluations and co-evaluations make $(n-1)\mathrm{Pr}_B$ dualizable. ■

Remark 1.113. This may be false for $n = 0$: indeed, we are asking for a countably presented ring to be a finitely presented modules. (A dualizable module is a finitely presented module.)

1.4.3 Descent

We are now ready to prove something a little harder.

Theorem 1.114 (Descent)**.** For each $n \geq 0$, the functor from $\mathbb{E}_\infty$-rings to $1\mathrm{Pr}$ defined by $A \mapsto n\mathrm{Pr}_A$ is a hypersheaf for the fpqc topology.

Remark 1.115. One could also try to get a hypersheaf to $\mathrm{CAlg}((n-1)\mathrm{Pr})$. This turns out to be the same because the forgetful functor preserves limits.

We should probably define some of these words. Let's start with fpqc.

Definition 1.116 (faithfully flat)**.** A map $f\colon A \to B$ of $\mathbb{E}_\infty$-rings is *faithfully flat* if and only if

- $\pi_0 f$ is faithfully flat, and
- the induced map $\pi_i A \otimes_{\pi_0 A} \pi_0 B \cong \pi_i B$ is an isomorphism for all $i \in \mathbb{Z}$.

The following explains why we will take fpqc to basically mean faithfully flat.

Lemma 1.117. Let $f\colon A \to B$ be a faithfully flat map of $\mathbb{E}_\infty$-rings. Then f is an $\aleph_1$-filtered colimit of countable presented faithfully flat maps $A \to B_i$.

Proof. Let's wave our hands. For classical rings, this is due to Waterhouse. One starts by doing this for modules, and then one uses the module construction to build some algebras. ■

Let's go ahead and engage with Theorem 1.114.

Proof of Theorem 1.114. By Lemma 1.111, $A \mapsto n\mathrm{Pr}_A$ commutes with finite products. Thus, it is enough to check that any fpqc hypercover $A \to A_\bullet$ induces an isomorphism

$$n\mathrm{Pr}_A \to \lim_{\Delta} n\mathrm{Pr}_{A_\bullet}$$

in $1\mathrm{Pr}$. By an elaboration of Lemma 1.117, every hypercover is an $\aleph_1$-filtered colimit of hypercovers which are level-wise countably presented $\mathbb{E}_\infty$-A-algebras. Thus, we may assume that each $A_\bullet$ in our cosimplicial object is a countably presented A-algebra.

As usual, we will induct. Let's start with $n = 0$. We want to show that the natural map

$$D(A) \to \lim_{\Delta} D(A_\bullet)$$

is an equivalence. The internal maps in the limit are given by f^*s. Because each $A_\bullet$ is countably presented, $D(A_\bullet)$ is self-dual by Lemma 1.112, so it is enough to show the dual of this map is an isomorphism. Namely, we want to show that the natural map

$$\operatorname*{colim}_{\Delta} D(A_\bullet) \to D(A)$$

is an isomorphism, where the colimit is taken over f_*s. Now, $f^!$ is the right adjoint of f_*, so it is enough to show that the natural map

$$D(A) \to \operatorname*{colim}_{\Delta} D(A_\bullet)$$

is an isomorphism, where the limit is taken over $f^!$s. Namely, $\operatorname{colim}^*_\Delta D(A_\bullet) = \lim^!_\Delta D(A_\bullet)$.

One now shows that this functor is fully faithful and essentially surjective separately. We will only discuss that it is fully faithful. Well, it admits a left adjoint given by the geometric realization of restriction of scalars along $D(A_\bullet) \to D(A)$. By a unit argument, it is equivalent to checking that the induced map

$$\operatorname*{colim}_{\Delta^{\mathrm{op}}} \mathrm{Hom}_A(A_\bullet, M) \to M$$

is an isomorphism. By taking $\mathrm{Hom}(-, M)$, it is enough to show that A is isomorphic to $\{\mathrm{Tot}^m A_\bullet\}_m$ as pro-objects, which turns out to follow from the hypersheaf condition: the cofiber of $A \to \mathrm{Tot}^m A_\bullet$ is ghost, and the composite of two ghost maps is null.

For the induction, we require the following.

Theorem 1.118 (Ambidexterity)**.** Let $f\colon A \to B$ be a countably presented map of $\mathbb{E}_\infty$-algebras. For $n \geq 1$, the functor $f_n^*\colon n\mathrm{Pr}_A \to n\mathrm{Pr}_B$ admits an $n\mathrm{Pr}_A$-linear left adjoint, which is isomorphic to f_{n*}.

The induction is now formal. Due to time, we omit details. ■

Let's discuss how one can produce the ambidexterity of Theorem 1.118.

Proposition 1.119. Let $\mathbb{C}$ be an $(\infty, 3)$-category, and let $F\colon X \to Y$ be a morphism in $\mathbb{C}$. Then the following are equivalent:

(i) F admits a right adjoint $F^R\colon Y \to X$, and $\mathrm{id}_X \to F^R F$ admits a right adjoint in $\mathrm{End}_{\mathbb{C}}(X)$, and $FF^R \to \mathrm{id}_Y$ admits a right adjoint in $\mathrm{End}_{\mathbb{C}}(Y)$; and

(ii) F admits a left adjoint $F^L\colon Y \to X$, and $\mathrm{id}_X \to F^R F$ admits a left adjoint in $\mathrm{End}_{\mathbb{C}}(X)$, and $FF^R \to \mathrm{id}_Y$ admits a left adjoint in $\mathrm{End}_{\mathbb{C}}(Y)$.

In this case, $F^L \cong F^R$.

Proof. Formal nonsense. The idea is that units becomes counits. According to Professor Hahn, this is a Five lemma for $(\infty,3)$-categories. ■

Proof of Theorem 1.118. We use Proposition 1.119. For $n \geq 1$, $\mathbb{C} \coloneqq (n+1)\mathrm{Pr}_A$ is an $(\infty,3)$-category. Then we will check (a) for the functor $f_n^* \colon n\mathrm{Pr}_A \to n\mathrm{Pr}_A$. We already know this has a right adjoint, and then we need to exhibit some more right adjoints, which is the purpose of countability hypotheses. ■

1.5 October 8

Today, we continue talking about quasicoherent sheaves.

1.5.1 Six Functors for Stacks

We now pass to stacks.

Definition 1.120 (stack)**.** A *stack* is an fpqc sheaf from $\mathbb{E}_\infty$-rings to spaces.

Definition 1.121. Fix a stack X, and let $\mathcal{U} \to X$ be an affine fpqc cover. Then we define $n\mathrm{Pr}_X$ to be the limit of the $n\mathrm{Pr}_U$ as U varies over $U \in \mathcal{U}$.

Remark 1.122. This is distinct from trying to define $n\mathrm{Pr}_X$ inductively from $D(X)$. Intuitively, the reason is that $D(X)$ is probably unable to remember the full content of the ∞-stack X, but the Stefanich ring should.

Remark 1.123. Here is another way to state this definition: the functor $n\mathrm{Pr}_\bullet$ on stacks is defined as the Kan extension of the functor $n\mathrm{Pr}_\bullet \colon \mathbb{E}_\infty\mathrm{Ring}^{\mathrm{op}} \to (n+1)\mathrm{Pr}$. This Kan extension takes limits with the underlying $(\infty,1)$-category structure, even though $(n+1)\mathrm{Pr}$ frequently admits module structure allowing it to pretend to be some weirder $(\infty,?)$-category.

To do descent, we need some countability hypothesis.

Definition 1.124 (bicountably presented)**.** A morphism $f \colon Y \to X$ of stacks is *bicountably presented* if and only if the following holds: for any pullback square

$$\begin{array}{ccc} \mathrm{Spec}\, A \times_X Y & \longrightarrow & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}\, A & \longrightarrow & X \end{array}$$

the map f' is a countable colimit of countably presented A-algebras. We let E be the subcategory of bicountably presented morphisms.

Remark 1.125. Tomer suspects that being countably presented can be detected on π_*, so these notions admit classical (graded) counterparts.

The goal of this lecture is to prove the following theorem.

Theorem 1.126. Let $f\colon Y \to X$ be a bicountably presented morphism of stacks. Choose $n \geq 1$.

(a) The functor $f_n^*\colon n\mathrm{Pr}_X \to n\mathrm{Pr}_Y$ admits a left adjoint $f_{n\sharp}$. This map lives in $(n+1)\mathrm{Pr}_X$ and commutes with base-change.

(b) Dualizability: if $n \geq 2$, then $f_{n\sharp}(1)$ is dualizable in $n\mathrm{Pr}_X$.

(c) Ambidexterity: if $n \geq 2$, then $f_{n\sharp} = f_{n*}$.

Intuitively, we are saying that (almost) all maps are étale for $n \geq 1$, and they are proper for $n \geq 2$. For example, for finite étale maps of schemes then f_* is a left adjoint of f^*. In general, an étale map should have $f^* = f^!$, which then has left adjoint $f_!$; there is a similar discussion for proper maps because $f_! = f_*$ for proper maps.

Remark 1.127. The affine case does not require $n \geq 2$, which is perhaps remarkable.

Proof of Theorem 1.126. Let's start with the case where $X = \operatorname{Spec} A$. We show (a) and (b).

(a) By hypothesis on f, we may write $Y = \operatorname{colim}_I \operatorname{Spec} B_i$ where I is countable, and each map $A \to B_i$ is countably generated. Then

$$n\mathrm{Pr}_Y = \lim_{I^*} n\mathrm{Pr}_{B_i},$$

where $I*$ means that we are taking the limit over the pullback morphisms. We already have a left adjoint for morphisms of the affine schemes, so

$$n\mathrm{Pr}_Y = \operatorname*{colim}_{I_\sharp} n\mathrm{Pr}_{B_i}.$$

There are now natural maps $f_{i,n\sharp}\colon n\mathrm{Pr}_{B_i} \to n\mathrm{Pr}_A$, and they assemble into the desired left adjoint $f_{n\sharp}\colon n\mathrm{Pr}_Y \to n\mathrm{Pr}_A$.

(b) Recall that $I_* = I_\sharp$. Thus, $f_{n\sharp}(1)$ is isomorphic to $\operatorname{colim}_{I_\sharp} f_{n\sharp}(1)$. We now conclude because a countable colimit of dualizable objects in $2\mathrm{Pr}$ along left adjoints is again dualizable. (This argument does not need $n \geq 2$.) Observe that there is something tricky happening here:

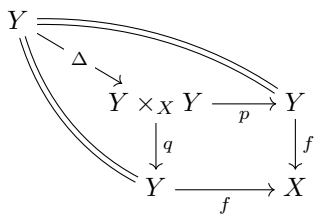
Let's now work globally with general X. As before, we can write

$$(n+1)\mathrm{Pr}_X = \lim_{\operatorname{Spec} A \to X} (n+1)\mathrm{Pr}_A.$$

Here, the limit may as well be taken in $1\mathrm{Pr}$. For (a), note that each map $n\mathrm{Pr}_Y \to n\mathrm{Pr}_A$ already has the desired left adjoint, and it turns out that compatibility with base-change allows these maps to glue to a map $n\mathrm{Pr}_Y \to n\mathrm{Pr}_X$. For (b), we don't say anything other than asserting that dualizability can be checked locally.[2]

It remains to handle ambidexterity. We will use Proposition 1.119. Thus, we want to show that the natural maps $\mathrm{id} \to f_n^* f_{n\sharp}$ and $f_{n\sharp} f_n^* \to \mathrm{id}$ have right adjoints.

- For the counit $f_{n\sharp} f_n^* \to \mathrm{id}$, by base-change, it is enough to just exhibit a right adjoint for $f_{n\sharp}(1) \to 1$. By some formula for dualizability, it is equivalent to ask for $1 \to f_{n*}1$ to have a left adjoint. This latter map is simply $f_{n-1}^*\colon (n-1)\mathrm{Pr}_X \to (n-1)\mathrm{Pr}_Y$, so we get the result by the existence of $f_{n-1,\sharp}$. Note that we have used $n \geq 2$ so that $n - 1 \geq 1$.
- For the unit $\mathrm{id} \to f_n^* f_{n\sharp}$, we build the following diagonal diagram.



[2] I am guessing that one can glue together a dual object.

Let the left downward projection be p. Now, $\mathrm{id} = p_{n\sharp}\Delta_{n\sharp}\Delta_n^* q_n^*$, and we want a map to $p_{n\sharp}q_n^* = f_n^* f_{n\sharp}$. (This second equality holds by base-change.) The desired map turns to be the map we constructed in the counit point for the morphism Δ! ■

Remark 1.128. The proof of (c) not only shows that $f_{n\sharp}$ and f_{n*} are the same, but also the identification is compatible with base-change. One could attempt to give an identification by building up from affines, but then it is not obvious that this construction is compatible with base-change.

Example 1.129. Another way to state (b) is that $(n-1)\mathrm{Pr}_Y$ is dualizable in $n\mathrm{Pr}_X$.

1.5.2 Applications

Let's apply Theorem 1.126 to show some things.

Corollary 1.130 (Künneth)**.** Fix bicountably presented morphisms $f\colon Y \to X$ and $f'\colon Y' \to X$. For any $n \geq 2$,

$$(n-1)\mathrm{Pr}_{Y\times_X Y'} \cong (n-1)\mathrm{Pr}_Y \otimes (n-1)\mathrm{Pr}_{Y'},$$

where the isomorphism (and tensor product) takes place in $n\mathrm{Pr}_X$.

Proof. By Theorem 1.126, we have (proper) base-change for these morphisms. Let's see this. We consider the following diagram.

$$\begin{array}{ccc} Y\times_X Y' & \xrightarrow{f} & Y' \\ {\scriptstyle f'}\downarrow & \searrow^{d} & \downarrow{\scriptstyle p'} \\ Y & \xrightarrow[p]{} & X \end{array}$$

Now, using the projection formula and base-change,

$$\begin{aligned} (n-1)\mathrm{Pr}_{Y\times_X Y'} &= d_* 1_{Y\times_X Y'} \\ &\cong p'_* f_*(f^* 1_{Y'} \otimes f'^* 1_Y) \\ &\overset{*}{\cong} p'_*(1_{Y'} \otimes f_* f'^* 1_Y) \\ &\cong p'_*(1_{Y'} \otimes p'^* p_* 1_Y) \\ &\cong p'_*(1_{Y'} \otimes p_* 1_Y \\ &= (n-1)\mathrm{Pr}_Y \otimes (n-1)\mathrm{Pr}_{Y'}. \end{aligned}$$

Here, $\overset{*}{\cong}$ is an application of base-change, which needs the full force of Theorem 1.126. Roughly speaking, what is going on is that formally base-change holds for $(-)_\sharp$ via some checks doable with (a), and base-change for our morphisms holds for $(-)_*$ by using Theorem 1.126(c). ■

Corollary 1.131. Fix maps $f\colon Y \to X$ and $f'\colon Y' \to X$ of $\aleph_1$-compact stacks. For $n \geq 2$, the right adjoints f_{n*} of f_n^* commute with small colimits, tensor products, and pullbacks, and there is a Künneth formula.

Proof. Omitted. ■

Remark 1.132. Without the countability hypothesis, it may happen that f_{n*} is not in $(n+1)\mathrm{Pr}_X$, but it will live in $\mathrm{Mod}_{n\mathrm{Pr}_X}\mathrm{Pr}^L$. Recall that these are different by Remark 1.122.

Corollary 1.133. There is a functor from $\aleph_1$-compact stacks to the opposite category of Stefanich rings given by sending X to

$$(\mathcal{O}(X), D(X), 1\mathrm{Pr}_X, 2\mathrm{Pr}_X, \ldots),$$

and it commutes with countable colimits and finite limits.

Proof. The construction allows us to see that we have built a Stefanich ring, and it commutes with colimits by construction of the $n\mathrm{Pr}_X$s. Thus, we will focus our energies on showing that it commutes with finite limits. It does send the final object $\mathrm{Spec}\,\mathbb{S}$ (which is the stack associated to $\mathbb{S}$) to the final opposite Stefanich ring $\mathbb{S}$.

It remains to show that our functor preserves fiber products. Roughly speaking, one expects the fiber product of $Y \to X$ and $Y' \to X$ to be a Stefanich ring which looks like

$$(\mathcal{O}(Y) \otimes_{\mathcal{O}(X)} \mathcal{O}(Y'), D(Y) \otimes_{D(X)} D(Y'), 1\mathrm{Pr}_Y \otimes_{1\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \ldots),$$

but this is not actually a Stefanich ring. However, there is a process by which one can force this to be a Stefanich ring, and it turns out to compare this to

$$\left(\mathcal{O}(Y) \otimes^{D(X)} \mathcal{O}(Y'), D(Y) \otimes^{1\mathrm{Pr}_X} D(Y'), 1\mathrm{Pr}_Y \otimes^{2\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \ldots\right),$$

which is also not literally a Stefanich ring (here the superscripts mean that the tensor products take place in that category), but it agrees with the correct fiber product Stefanich ring. ■

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List of Definitions

atomic, 15
atomically generated, 15

bicountably presented, 21

compact, 7
compactly assembled, 13

dualizable, 11

faithfully flat, 19
filtered, 7

internal left adjoint, 15

Lurie tensor product, 10

presentable, 8

regular, 7

stable, 12
stack, 21
Stefanich ring, 6